

Problem Set 1

Probability and Statistical Inference

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1. **Identification and estimation.** A voluntary study-skills program is evaluated using final examination scores. Let $D_i = 1$ if student i participates in the program and $D_i = 0$ otherwise. Let Y_i be the observed score. The sample is

D_i	1	1	1	1	0	0	0	0
Y_i	78	74	82	86	65	72	68	71

Let

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0, \quad \beta = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0].$$

- (a) Compute $n_1, n_0, \bar{Y}_1, \bar{Y}_0$ and $\hat{\beta}$. Then identify the estimator, its numerical estimate in this sample, and the parameter.
- (b) Introduce potential outcomes $Y(1)$ and $Y(0)$ and write the equation that links them to the observed outcome Y . Define the ATET.
- (c) Show that

$$\beta = \text{ATET} + [\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]].$$

Give an interpretation of the term in square brackets.

- (d) In the population, suppose

$$\mathbb{E}[Y(1) \mid D = 1] = 78, \quad \mathbb{E}[Y(0) \mid D = 1] = 72, \quad \mathbb{E}[Y(0) \mid D = 0] = 68.$$

Compute β , the selection-bias term, and the ATET. Does the raw population difference overstate or understate the ATET?

- (e) Now suppose that program participation were randomly assigned, so that $D \perp\!\!\!\perp (Y(0), Y(1))$. What additional conclusions would follow? Explain why a very large sample, by itself, cannot correct a failure of this identification assumption.

2. **Probability rules and updating.** You may use the probability axioms: non-negativity, $\mathbb{P}(\Omega) = 1$, and countable additivity for disjoint events.

- (a) Prove that $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- (b) If $A \subseteq B$, prove that $\mathbb{P}(A) \leq \mathbb{P}(B)$.
- (c) Show that $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$.
- (d) A data-quality problem is present with prior probability 0.30. An automatic check raises a warning with probability 0.80 when the problem is present and with probability 0.20 when it is absent. Find the probability that the problem is present after a warning is observed. Identify the prior, likelihood and posterior in your calculation, and explain why $\mathbb{P}(\text{warning} \mid \text{problem})$ is not the requested posterior.

3. **An exact sampling distribution.** Each observation is equally likely to be 0 or 2, and the population mean is $\theta = 1$. For a sample of size two, the four ordered samples (0, 0), (0, 2), (2, 0), (2, 2) each have probability 1/4. Consider $T_2 = (X_1 + X_2)/2$ and $S_2 = X_1$.

- (a) Calculate both estimators in each possible sample. Hence give their sampling distributions: their possible values and the probability of each value.
- (b) You are given that both sampling means equal 1, while $\text{Var}(T_2) = 1/2$ and $\text{Var}(S_2) = 1$. Calculate their biases and MSEs. Which is more efficient within this pair of unbiased estimators?
- (c) For independent samples of size n , define $T_n = \bar{X}_n$ and $S_n = X_1$. Their sampling means and variances are given below:

	mean	variance
T_n	1	$1/n$
S_n	1	1

Describe what happens to the centre and spread of each sampling distribution as n grows. Explain directly why S_n is not consistent for $\theta = 1$.

- (d) Explain the difference between increasing the sample size n and increasing the number R of repeated samples used to draw a histogram of T_n . Which one changes the sampling distribution of the estimator?

4. **MSE and consistency.** Use Lecture 1's identity $\text{MSE} = \text{Bias}^2 + \text{Var}$ and its definition of consistency.

- (a) An estimator T equals $\theta - 1$ or $\theta + 3$, each with probability 1/2. Its sampling mean is $\theta + 1$ and its variance is 4. Calculate its bias and MSE. Check the MSE by averaging the squared errors in these two equally likely cases.
- (b) An estimator V_n equals $\theta - 1/n$ or $\theta + 1/n$, each with probability 1/2. Its sampling mean is θ and its variance is $1/n^2$. Calculate its bias and MSE. Then use the definition of consistency directly to show that $V_n \xrightarrow{p} \theta$.
- (c) Let $W_n = n$ with probability $1/n$ and $W_n = 0$ otherwise, viewed as an estimator of zero. You are given that its bias is 1 and its variance is $n - 1$. Show directly that $W_n \xrightarrow{p} 0$. What does this example tell us about whether consistency requires bias and variance to converge to zero?

5. **Four estimators.** Consider the following version of Lecture 1's normal sampling examples:

$$A_n = X_1, \quad B_n = \bar{X}_n, \quad C_n = \bar{X}_n + \frac{2}{n}, \quad D_n = 0.$$

Their sampling distributions are given; you do not need to derive them. Here $N(m, v)$ denotes a normal distribution with mean m and variance v .

estimator	sampling distribution
A_n	$N(\theta, 4)$
B_n	$N(\theta, 4/n)$
C_n	$N(\theta + 2/n, 4/n)$
D_n	always equals 0

- For each estimator, calculate its bias, variance and MSE.
- Which estimators are unbiased, and which are consistent for a general unknown θ ? Explain consistency using the centres and spreads of the normal curves, as in Lecture 1.
- Compare A_n and B_n within the class of unbiased estimators. Why does D_n show that smallest variance is not a useful definition of efficiency unless the target and the set of estimators being compared are specified?
- Which estimator illustrates that an estimator may be biased at every finite n and nevertheless be consistent?

6. **Sample summaries and interpretation.** Consider the ordered sample

$$1, 2, 3, 4, 5, 6, 14.$$

Use the lecture convention $\hat{q}_p = X_{(\lceil np \rceil)}$.

- Compute the sample mean, median, first and third quartiles, and interquartile range.
- Replace the largest observation, 14, by 70 and recompute these summaries. Which measures are most sensitive to this change?
- State whether each claim is true or false and explain briefly.
 - Increasing the number of simulation repetitions R reduces the variance of the estimator whose sampling distribution is being simulated.
 - An estimator can be biased for every finite sample size and still be consistent.
 - An estimator can have small MSE even though its realized estimate in one particular sample is far from the target.
 - If $\hat{\beta}$ estimates β extremely precisely, then β must equal the causal effect of interest.